CSL 7640: Natural Language Understanding - Assignment 2

Word Embeddings and RNN-based Name Generation

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1 Problem 1: Word Embeddings from IIT Jodhpur Data

1.1 Dataset Preparation

1.1.1 Collection and Preprocessing

- Official website pages covering departments, academic programs, and research initiatives
- Academic regulation and policy documents
- Faculty profile pages and their research descriptions
- Course syllabus and course descriptions

The raw collected text underwent comprehensive preprocessing to create a clean, machine-readable corpus. The preprocessing pipeline included:

1. **Boilerplate Removal:** Elimination of navigation elements, HTML artifacts, and repetitive institutional text that does not contribute to semantic understanding
2. **Tokenization:** Breaking down text into individual words and meaningful units
3. **Lowercasing:** Converting all text to lowercase to treat different capitalizations of the same word uniformly
4. **Punctuation Handling:** Removal of excessive punctuation and special characters while preserving semantic integrity
5. **Language Filtering:** Ensuring only English text was retained, removing any non-English content

1.1.2 Dataset Statistics

Metric	Value
Documents	200
Total Tokens	71,544
Vocabulary Size	6,628

Table 1: Dataset Statistics

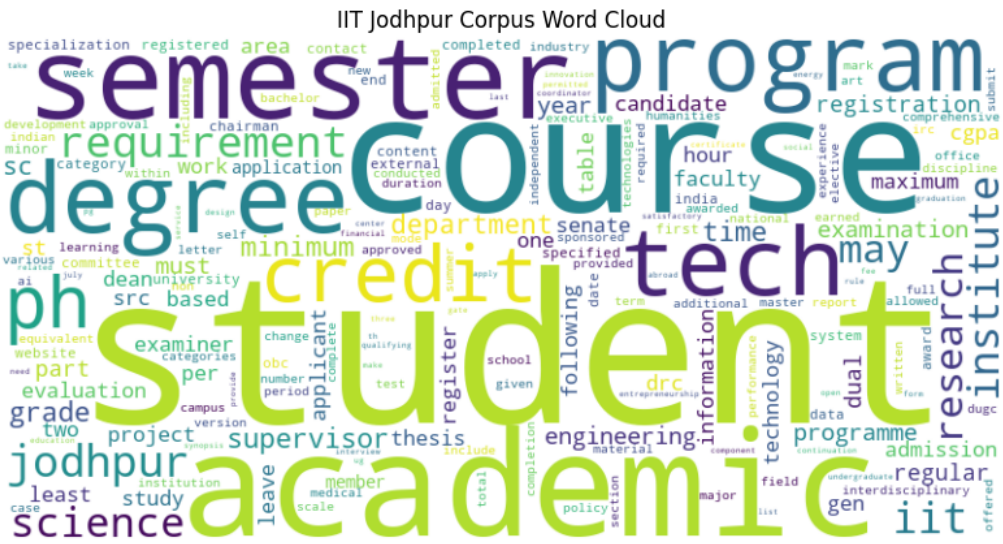


Figure 1: Word Cloud - Most Frequent Terms

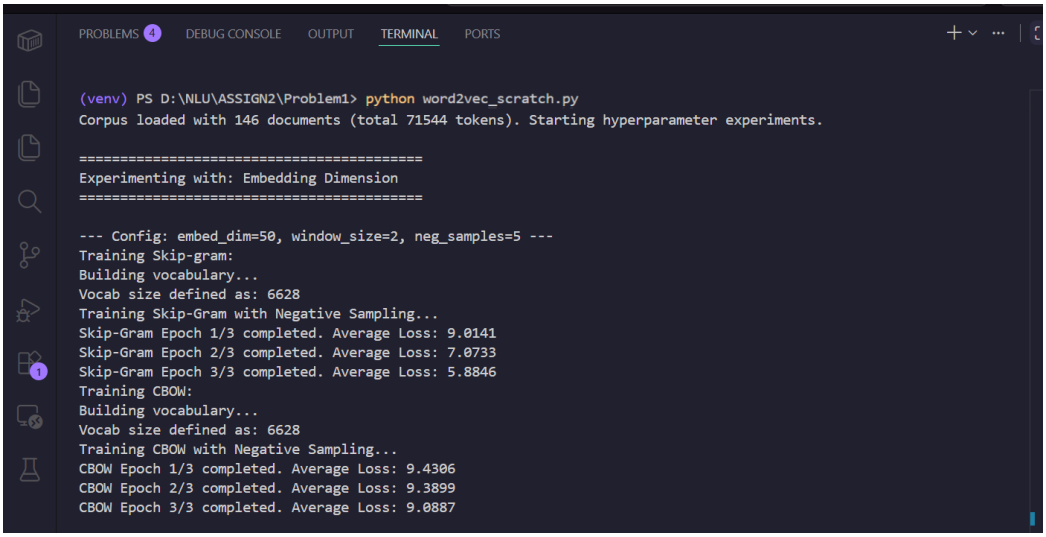
Dominant terms: **students, program, course, degree, research**, which are appropriate for academic documents.

1.2 Model Training and Hyperparameter Experiments

1.2.1 Architecture Overview

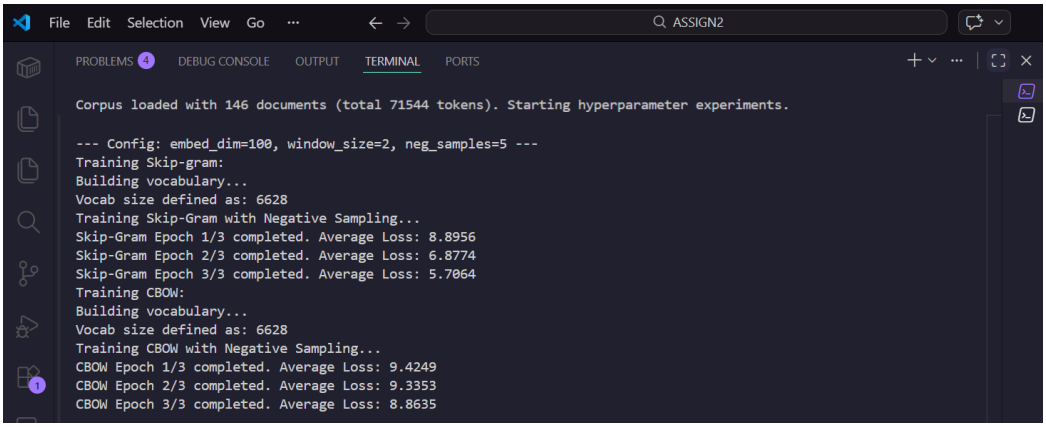
Skip-gram: Predicts context words from target word. Effective for semantic relationships and rare words.
CBOW: Predicts target word from context. Computationally efficient, captures co-occurrence patterns.
Both use negative sampling.
Hyperparameters Experiments: Embedding Dimension (50, 100, 150), Context Window (2, 4, 6), Negative Samples (3, 5, 10).
So, **9 combinations.**

1.2.2 Embedding Dimension Impact



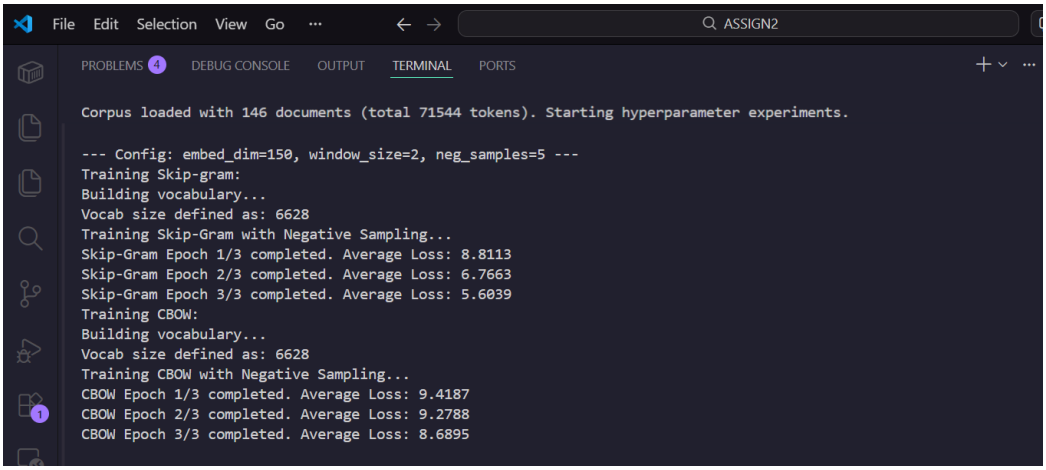
```
(venv) PS D:\NLU\ASSIGN2\Problem1> python word2vec_scratch.py
Corpus loaded with 146 documents (total 71544 tokens). Starting hyperparameter experiments.

==========================================
Experimenting with: Embedding Dimension
==========================================

--- Config: embed_dim=50, window_size=2, neg_samples=5 ---
Training Skip-gram:
Building vocabulary...
Vocab size defined as: 6628
Training Skip-Gram with Negative Sampling...
Skip-Gram Epoch 1/3 completed. Average Loss: 9.0141
Skip-Gram Epoch 2/3 completed. Average Loss: 7.0733
Skip-Gram Epoch 3/3 completed. Average Loss: 5.8846
Training CBOW:
Building vocabulary...
Vocab size defined as: 6628
Training CBOW with Negative Sampling...
CBOW Epoch 1/3 completed. Average Loss: 9.4306
CBOW Epoch 2/3 completed. Average Loss: 9.3899
CBOW Epoch 3/3 completed. Average Loss: 9.0887
```

(a) Dim=50



```
Corpus loaded with 146 documents (total 71544 tokens). Starting hyperparameter experiments.

--- Config: embed_dim=100, window_size=2, neg_samples=5 ---
Training Skip-gram:
Building vocabulary...
Vocab size defined as: 6628
Training Skip-Gram with Negative Sampling...
Skip-Gram Epoch 1/3 completed. Average Loss: 8.8956
Skip-Gram Epoch 2/3 completed. Average Loss: 6.8774
Skip-Gram Epoch 3/3 completed. Average Loss: 5.7064
Training CBOW:
Building vocabulary...
Vocab size defined as: 6628
Training CBOW with Negative Sampling...
CBOW Epoch 1/3 completed. Average Loss: 9.4249
CBOW Epoch 2/3 completed. Average Loss: 9.3353
CBOW Epoch 3/3 completed. Average Loss: 8.8635
```

(b) Dim=100



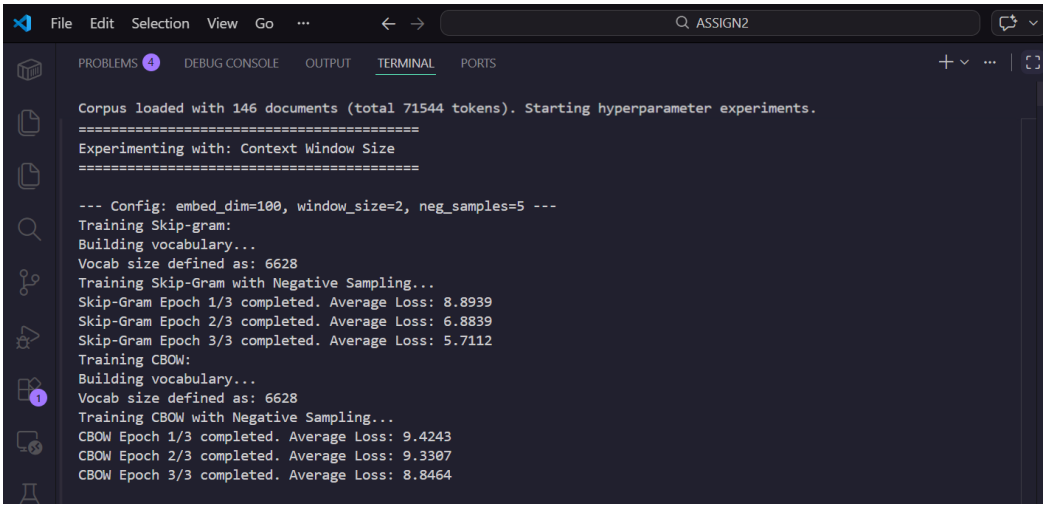
```
Corpus loaded with 146 documents (total 71544 tokens). Starting hyperparameter experiments.

--- Config: embed_dim=150, window_size=2, neg_samples=5 ---
Training Skip-gram:
Building vocabulary...
Vocab size defined as: 6628
Training Skip-Gram with Negative Sampling...
Skip-Gram Epoch 1/3 completed. Average Loss: 8.8113
Skip-Gram Epoch 2/3 completed. Average Loss: 6.7663
Skip-Gram Epoch 3/3 completed. Average Loss: 5.6039
Training CBOW:
Building vocabulary...
Vocab size defined as: 6628
Training CBOW with Negative Sampling...
CBOW Epoch 1/3 completed. Average Loss: 9.4187
CBOW Epoch 2/3 completed. Average Loss: 9.2788
CBOW Epoch 3/3 completed. Average Loss: 8.6895
```

(c) Dim=150

Figure 2: Embedding Dimension for Skip-gram: Loss 5.88→5.70→5.60
Embedding Dimension for CBOW: Loss 9.09→8.86→8.69

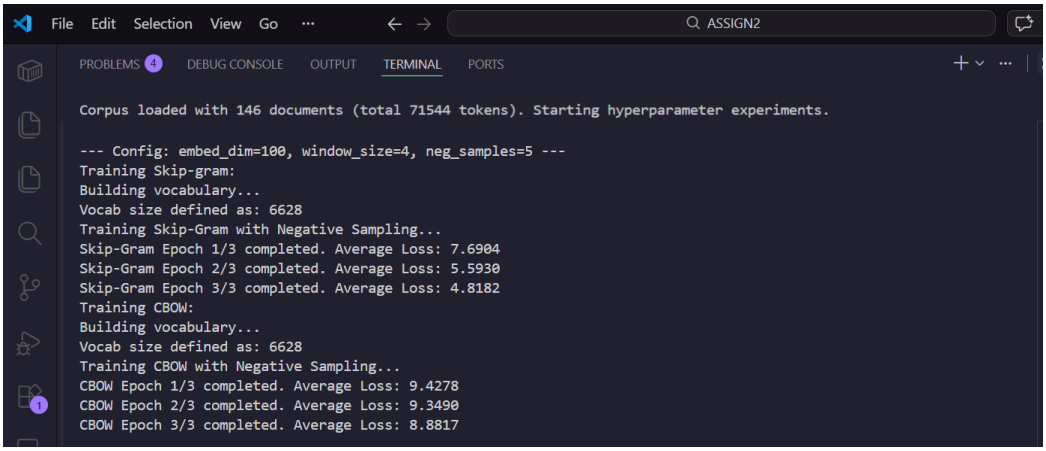
Skip-gram improves with larger dimensions; CBOW remains stable. Diminishing returns beyond 100.

1.2.3 Context Window Size Impact



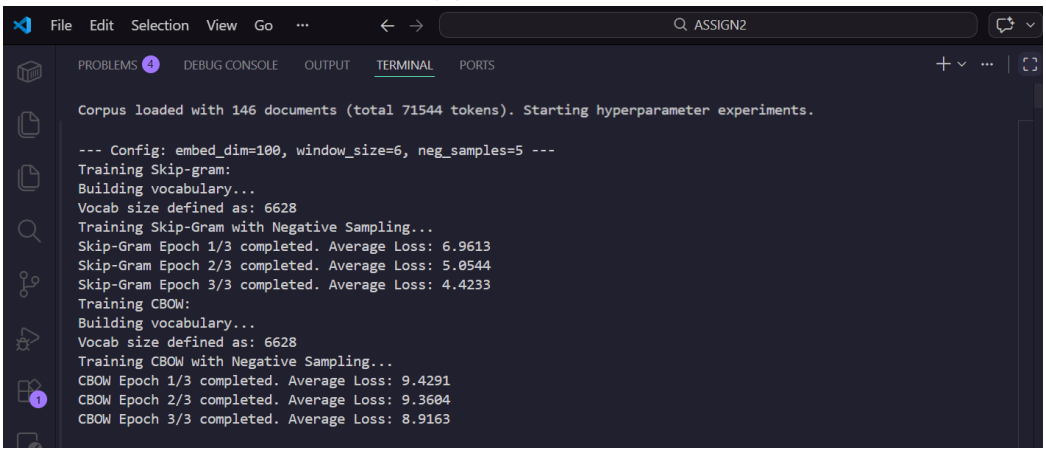
```
Corpus loaded with 146 documents (total 71544 tokens). Starting hyperparameter experiments.
========================================
Experimenting with: Context Window Size
========================================

--- Config: embed_dim=100, window_size=2, neg_samples=5 ---
Training Skip-gram:
Building vocabulary...
Vocab size defined as: 6628
Training Skip-Gram with Negative Sampling...
Skip-Gram Epoch 1/3 completed. Average Loss: 8.8939
Skip-Gram Epoch 2/3 completed. Average Loss: 6.8839
Skip-Gram Epoch 3/3 completed. Average Loss: 5.7112
Training CBOW:
Building vocabulary...
Vocab size defined as: 6628
Training CBOW with Negative Sampling...
CBOW Epoch 1/3 completed. Average Loss: 9.4243
CBOW Epoch 2/3 completed. Average Loss: 9.3307
CBOW Epoch 3/3 completed. Average Loss: 8.8464
```

(a) Window=2



```
Corpus loaded with 146 documents (total 71544 tokens). Starting hyperparameter experiments.

--- Config: embed_dim=100, window_size=4, neg_samples=5 ---
Training Skip-gram:
Building vocabulary...
Vocab size defined as: 6628
Training Skip-Gram with Negative Sampling...
Skip-Gram Epoch 1/3 completed. Average Loss: 7.6904
Skip-Gram Epoch 2/3 completed. Average Loss: 5.5930
Skip-Gram Epoch 3/3 completed. Average Loss: 4.8182
Training CBOW:
Building vocabulary...
Vocab size defined as: 6628
Training CBOW with Negative Sampling...
CBOW Epoch 1/3 completed. Average Loss: 9.4278
CBOW Epoch 2/3 completed. Average Loss: 9.3490
CBOW Epoch 3/3 completed. Average Loss: 8.8817
```

(b) Window=4



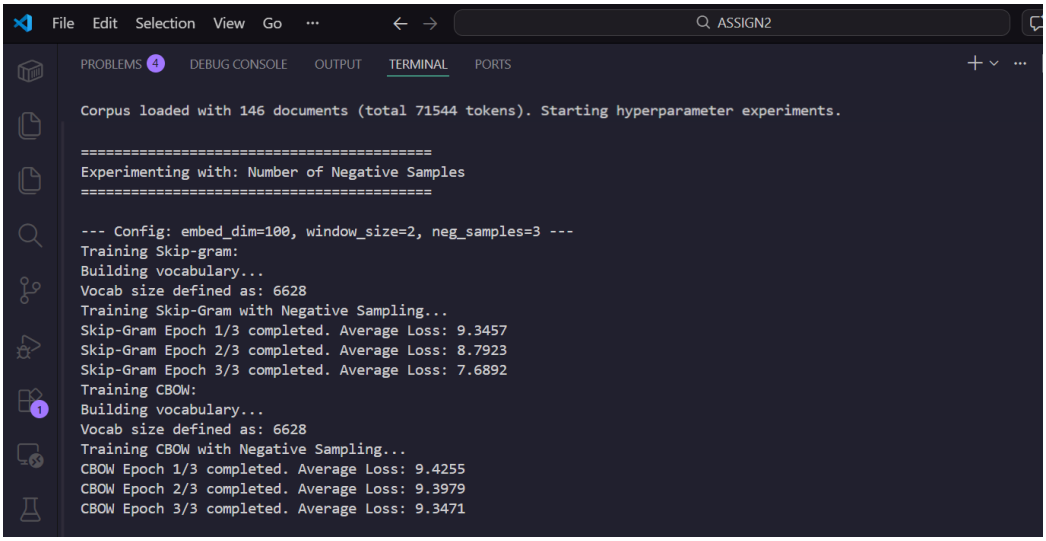
```
Corpus loaded with 146 documents (total 71544 tokens). Starting hyperparameter experiments.

--- Config: embed_dim=100, window_size=6, neg_samples=5 ---
Training Skip-gram:
Building vocabulary...
Vocab size defined as: 6628
Training Skip-Gram with Negative Sampling...
Skip-Gram Epoch 1/3 completed. Average Loss: 6.9613
Skip-Gram Epoch 2/3 completed. Average Loss: 5.0544
Skip-Gram Epoch 3/3 completed. Average Loss: 4.4233
Training CBOW:
Building vocabulary...
Vocab size defined as: 6628
Training CBOW with Negative Sampling...
CBOW Epoch 1/3 completed. Average Loss: 9.4291
CBOW Epoch 2/3 completed. Average Loss: 9.3604
CBOW Epoch 3/3 completed. Average Loss: 8.9163
```

(c) Window=6

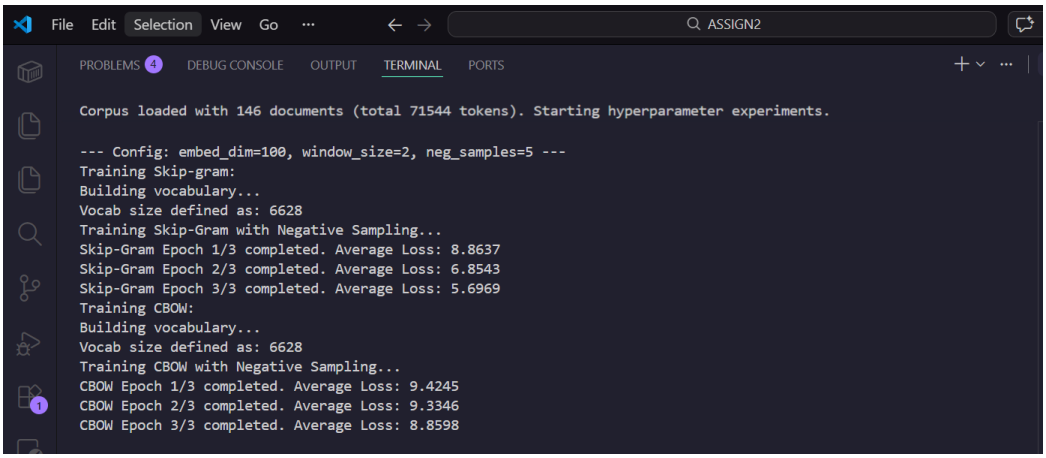
Figure 3: Window Size: Skip-gram Loss 5.71→4.82→4.42
Window Size: CBOW Loss stable 8.84-8.91

Larger windows significantly improve Skip-gram by capturing distant word relationships. CBOW insensitive to window size.

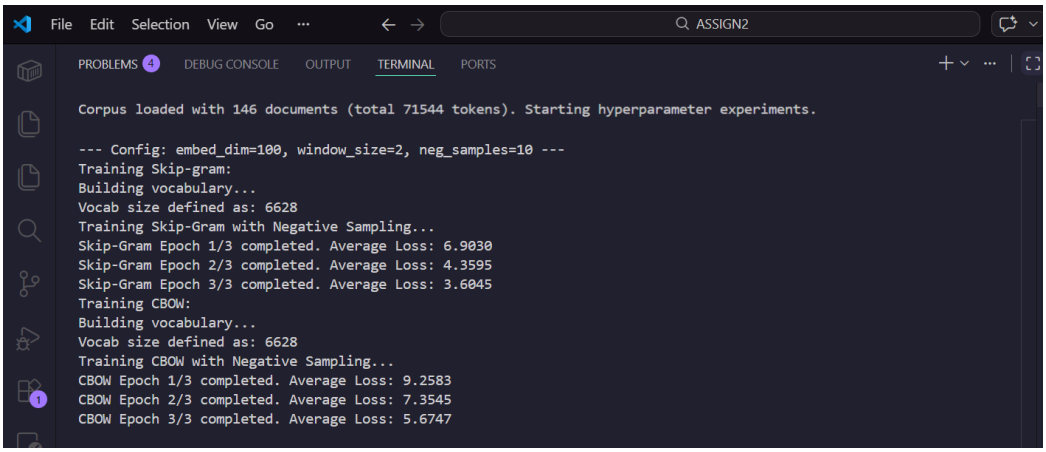
1.2.4 Negative Samples Impact



(a) Neg=3



(b) Neg=5



(c) Neg=10

Figure 4: Negative Samples: Skip-gram Loss 7.69→5.70→3.60
Negative Samples: CBOW Loss 9.35→8.86→5.67

Both models improve substantially with more negative samples. Diminishing returns beyond 10.

Dimension	Window size	Negative sampling	Skip-Gram Loss	CBOW Loss
50	2	5	5.8846	9.0887
100	2	5	5.7064	8.8635
150	2	5	5.6039	8.6895
100	2	5	5.7112	8.8464
100	4	5	4.8182	8.8817
100	6	5	4.4233	8.9163
100	2	3	7.6892	9.3471
100	2	5	5.6969	8.8598
100	2	10	3.6045	5.6747

Table 2: Hyperparameter Experiment Results

1.3 Semantic Analysis

1.3.1 Nearest Neighbors Analysis

	Skip-gram		CBOW	
Research	**Neighbors**	**Similarity Score**	**Neighbors**	**Similarity Score**
	conversant	0.762	students	0.957
	sub	0.7540	student	0.9501
	proposal	0.748	may	0.9468
	work	0.740	course	0.931
	areas	0.7333	semester	0.9301
Student	**Neighbors**	**Similarity Score**	**Neighbors**	**Similarity Score**
	semester	0.726	students	0.9780
	permitted	0.7256	semester	0.9578
	candidate	0.7183	program	0.957
	change	0.7118	academic	0.9568
	alive	0.6937	requirements	0.9526
PhD	**Neighbors**	**Similarity Score**	**Neighbors**	**Similarity Score**
	dual	0.9424	tech	0.652
	mtech	0.9378	program	0.6469
	advanced	0.9192	ph	0.6268
	regulations	0.9181	degree	0.6236
	manufacturing	0.9155	regular	0.6185
Exam	**Neighbors**	**Similarity Score**	**Neighbors**	**Similarity Score**
	mission	0.907	icia	0.5247
	day	0.906	certificates	0.498
	bypass	0.9026	competitiveness	0.4885
	act	0.8935	maintained	0.462
	attempt	0.8932	barbara	0.4559

Table 3: Top Neighbors (Cosine Similarity)

Skip-gram: Better semantic coherence. Identifies contextually relevant terms (research→proposal, phd→mtech).

CBOW: Strong co-occurrence patterns but less semantic precision. Plural/singular associations dominant.

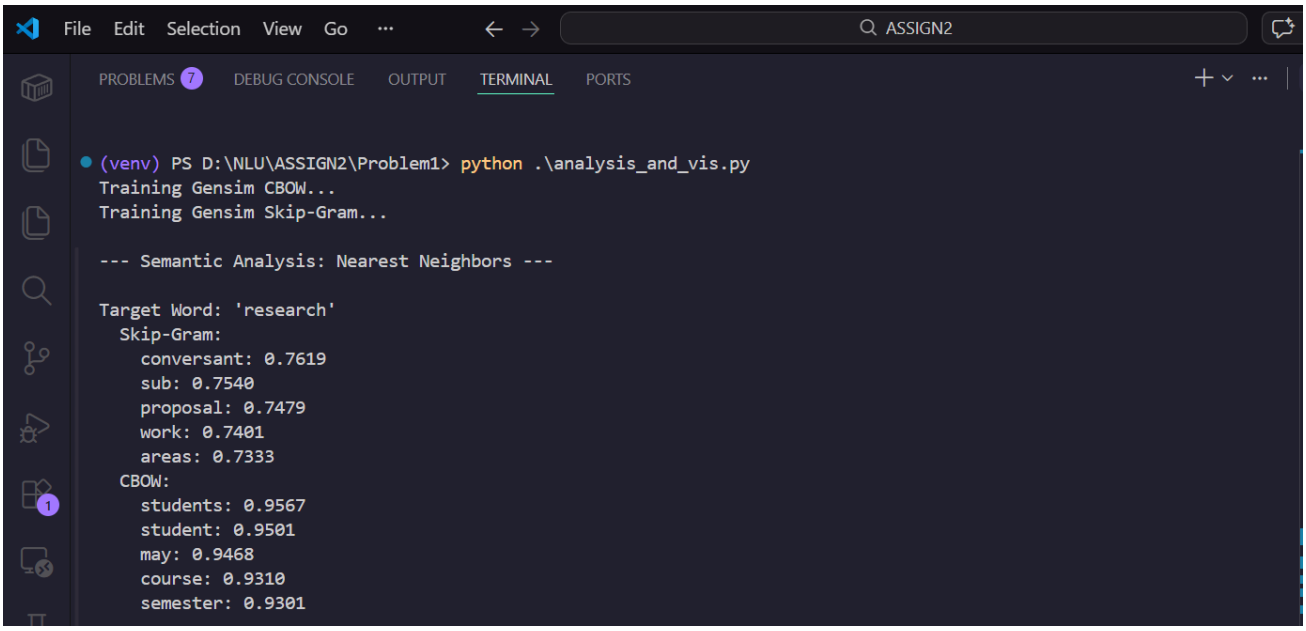


Figure 5: Nearest Neighbors Example 1

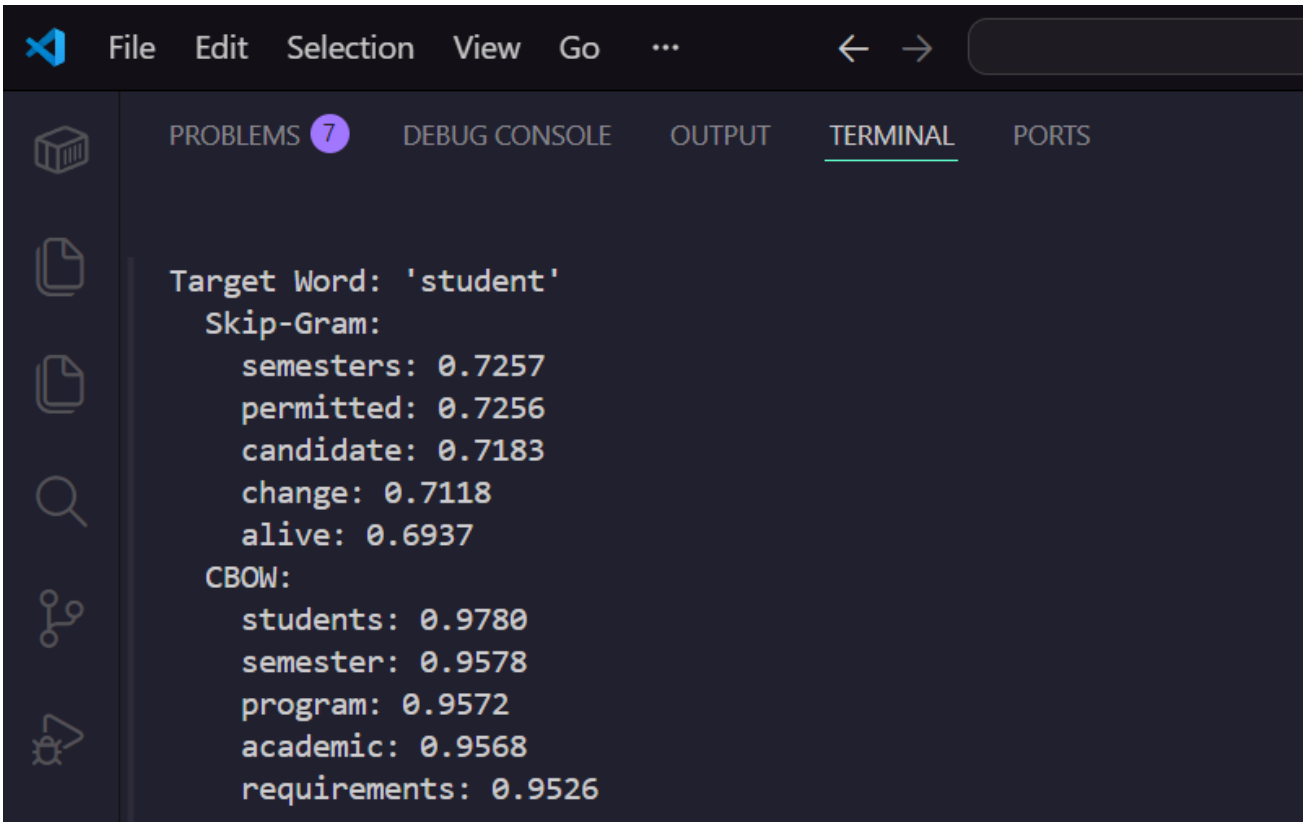


Figure 6: Nearest Neighbors Example 2

```
Target Word: 'phd'
  Skip-Gram:
    dual: 0.9424
    mtech: 0.9378
    advanced: 0.9192
    regulations: 0.9181
    manufacturing: 0.9155
  CBOW:
    tech: 0.6520
    program: 0.6469
    ph: 0.6268
    degree: 0.6236
    regular: 0.6185
```

Figure 7: Nearest Neighbors Example 3

```
Target Word: 'exam'
  Skip-Gram:
    mission: 0.9068
    day: 0.9064
    bypass: 0.9026
    act: 0.8935
    attempt: 0.8932
  CBOW:
    icia: 0.5247
    certificates: 0.4979
    competitiveness: 0.4885
    maintained: 0.4621
    barbara: 0.4559
```

Figure 8: Nearest Neighbors Example 4

1.3.2 Analogy Results

UG : BTech :: PG : ?		**Faculty : Teaching :: Student : ?**	
Skip-gram: master (0.58)	CBOW: dynamic (0.45)	Skip-gram: semesters (0.65)	CBOW: students (0.96)
Research : PhD :: Course : ?			
Skip-gram: courses (0.73), independent (0.72)		CBOW: courses (0.79), student (0.81)	

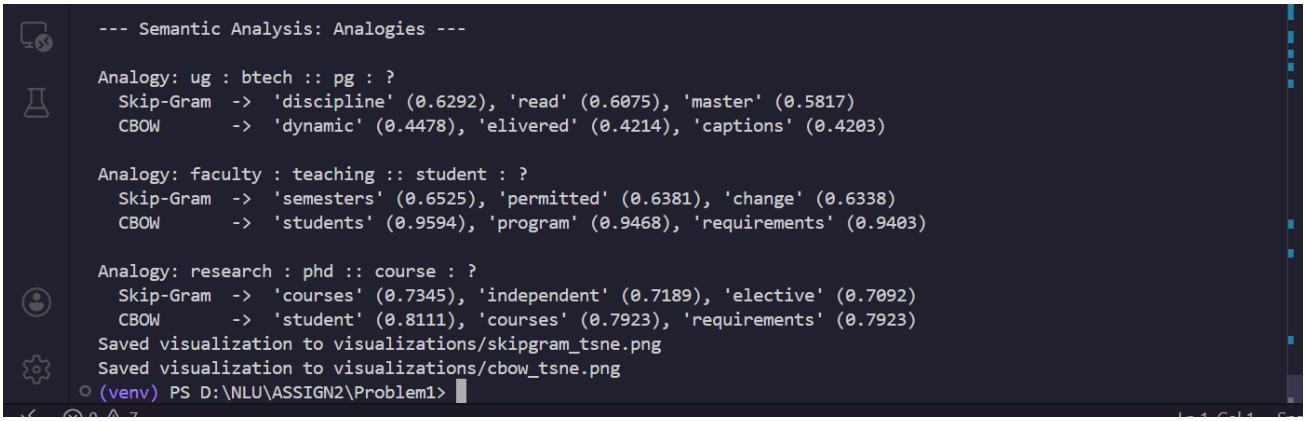
Table 4: Analogy Experiments

Best result: Research-PhD-Course analogy where both models identify "courses" correctly. Other analogies weak, indicating insufficient pattern exposure.

1.3.3 Summary of Semantic Analysis

The semantic analysis reveals important strengths and limitations:

- **Skip-gram Strengths:** Better at identifying contextually relevant and semantically coherent neighbors, particularly for less frequent words. It is giving nice performance on specific domain terms like "phd" and "research".
- **CBOW Strengths:** Captures strong co-occurrence patterns and maintains coherence for frequent words. Produces high similarity scores, indicating confident associations.
- **Limitations:** Analogy tasks reveal that both models struggle with capturing proportional semantic relationships, possibly due to limited corpus size and specific patterns. Some word pairs have insufficient context in the institutional documents.



```
--- Semantic Analysis: Analogies ---

Analogy: ug : btech :: pg : ?
  Skip-Gram  ->  'discipline' (0.6292), 'read' (0.6075), 'master' (0.5817)
  CBOW       ->  'dynamic' (0.4478), 'elivered' (0.4214), 'captions' (0.4203)

Analogy: faculty : teaching :: student : ?
  Skip-Gram  ->  'semesters' (0.6525), 'permitted' (0.6381), 'change' (0.6338)
  CBOW       ->  'students' (0.9594), 'program' (0.9468), 'requirements' (0.9403)

Analogy: research : phd :: course : ?
  Skip-Gram  ->  'courses' (0.7345), 'independent' (0.7189), 'elective' (0.7092)
  CBOW       ->  'student' (0.8111), 'courses' (0.7923), 'requirements' (0.7923)
Saved visualization to visualizations/skipgram_tsne.png
Saved visualization to visualizations/cbow_tsne.png
(venv) PS D:\NLU\ASSIGN2\Problem1>
```

Figure 9: Analogy Results Visualization

1.4 Clustering Visualization

1.4.1 Comparative Analysis: Skip-gram vs. CBOW

Aspect	Skip-gram	CBOW
Cluster Tightness	Tight, well-defined clusters	Larger, more diffuse clusters
Semantic Coherence	Higher (better semantic grouping)	Good (strong central nodes)
Rare Word Handling	Better representation	Weaker discrimination
Training Efficiency	Slower (predicts multiple outputs)	Faster (single output)
Convergence	Better with larger windows	Stable, insensitive to windows
Domain Specialization	Superior (discovers domain structure)	Adequate (captures patterns)

Table 5: Comparative Summary of Skip-gram vs. CBOW

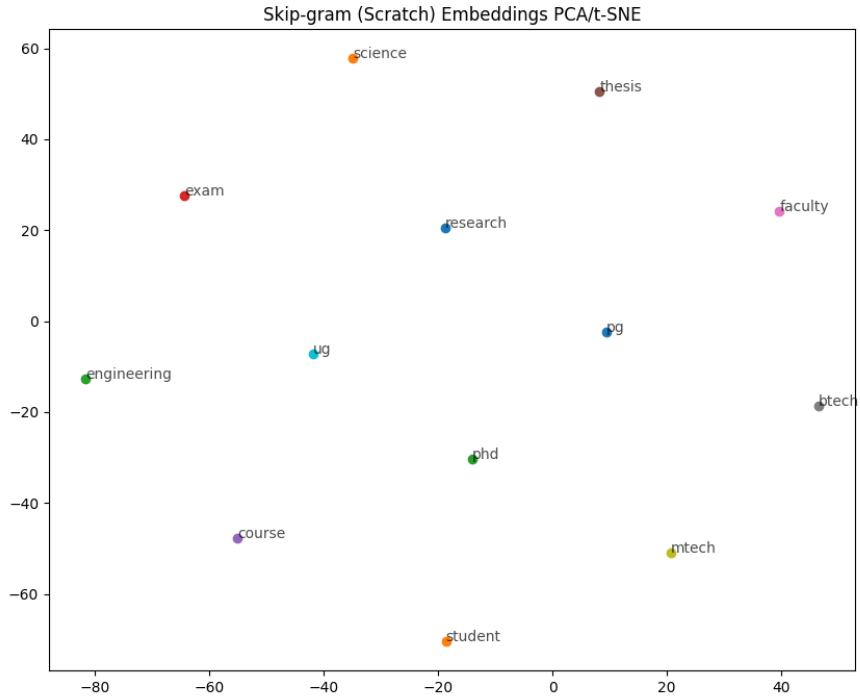


Figure 10: Skip-gram t-SNE: Tight degree clusters (btech, mtech, phd, dual), regulatory cluster (requirement, policy)

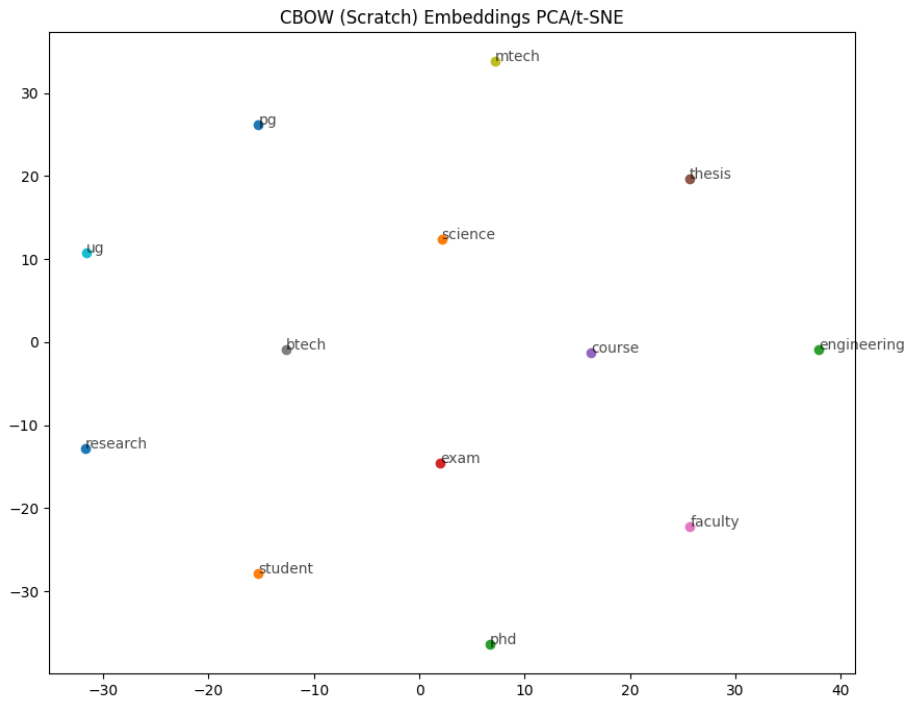


Figure 11: CBOW t-SNE: Larger diffuse clusters, strong central nodes (student, course, program)

Skip-gram advantages: Tighter semantic clusters, better domain specialization.
CBOW advantages: Faster training, strong co-occurrence capture, stable performance.

2 Problem 2: Character-Level Name Generation

2.1 Dataset and Models

Generated 1000 Indian names as training data. Implemented three architectures:

Model	Parameters	Hidden Size	Layers
Vanilla RNN	13,738	256	1
BLSTM	74,666	256	1 (bidir)
RNN + Attention	22,058	256	1

Table 6: Model Configurations

2.1.1 Architecture Details

Vanilla RNN: Single RNN layer with tanh activation. Baseline model.

$$h_t = \tanh(W_{hx}x_t + W_{hh}h_{t-1} + b_h)$$

BLSTM: Bidirectional LSTM processing sequences forward and backward. LSTM gating:

$$i_t = \sigma(W_{ii}x_t + W_{hi}h_{t-1}), \quad f_t = \sigma(W_{if}x_t + W_{hf}h_{t-1})$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_{ig}x_t + W_{hg}h_{t-1})$$

RNN + Attention: Vanilla RNN with attention over hidden states:

$$e_i = W_a \tanh(W_h h_i), \quad \alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}, \quad c_t = \sum_i \alpha_i h_i$$

2.2 Training and Evaluation

2.2.1 Training Curves

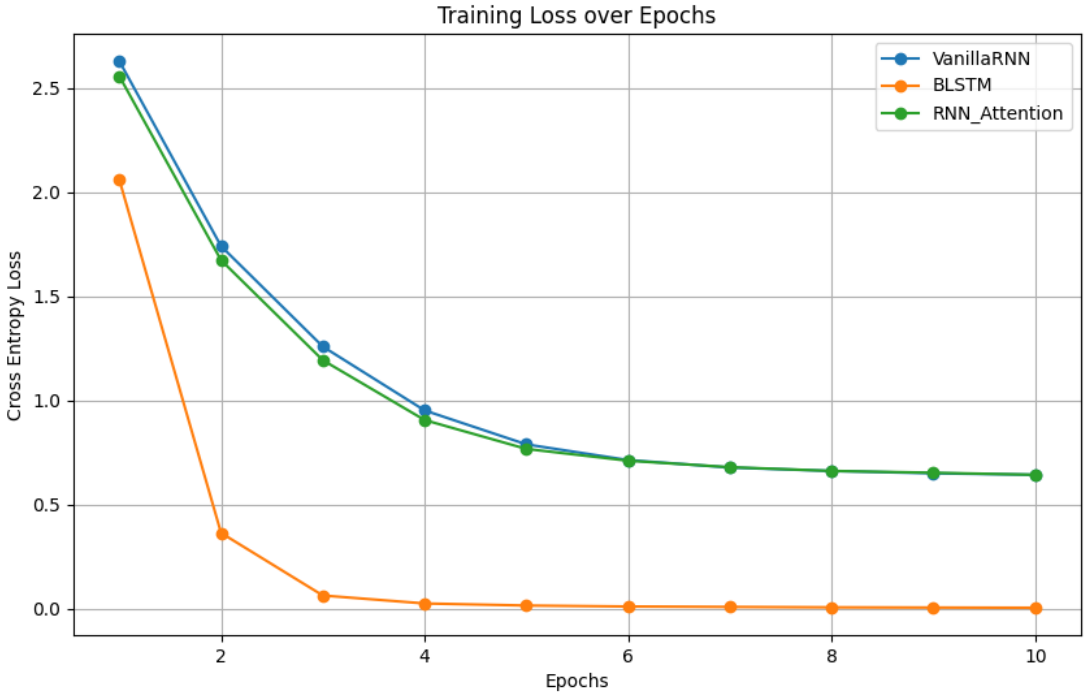


Figure 12: Training Loss Curves (10 epochs)

BLSTM: Rapid convergence 2.08→0.0042 (overfitting indicator).
Vanilla RNN: Steady convergence 2.64→0.64.
Attention-RNN: Moderate convergence 2.57→0.64.

2.2.2 Evaluation Metrics

Novelty: % of generated names not in training set.
Diversity: % of unique generated names.

Model	Diversity	Novelty
Vanilla RNN	93.00%	93.55%
BLSTM	100.00%	100.00%
RNN + Attention	93.00%	88.17%

Table 7: Quantitative Metrics (100 generated names)

BLSTM's perfect 100% scores are **meaningless**—they indicate severe overfitting. Perfect metrics mask catastrophic output quality failure (see qualitative analysis below).

2.3 Generated Samples

2.3.1 Vanilla RNN - BEST PRACTICAL RESULTS

```
Krishna Pandey
Nikhetha
Arjun Sidhtu
Priya Verma
Karan Pandey
```

OPTIMAL: Most realistic, phonetically plausible names. Proper structure, minimal repetition.

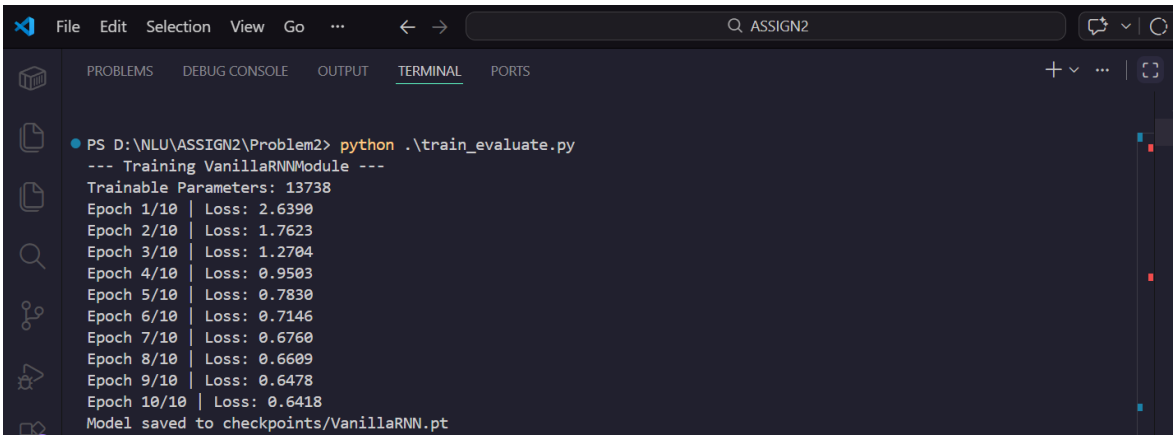


Figure 13: Vanilla RNN Loss

2.3.2 BLSTM - OVERFITTED (Poor Quality)

```
AKral A Agang AdIyJ
Agjit AgaxAng Aggta
AabAma Arya ASvit Ag
Pataj Poow PPPa Pata
```

FAILURE: Excessive character repetition (AA, PPP), gibberish sequences, unnatural patterns. Despite perfect novelty/diversity scores, output is unusable. Bidirectional processing inappropriate for unidirectional generation; model memorized noise patterns.

Figure 14: BLSTM Loss

2.3.3 RNN + Attention

```
Kajri
Senanya Mukherjee
Sindha
Arjun Verma
Reyansh
```

SUB-OPTIMAL: Attention mechanism prevents overfitting while maintaining good quantitative metrics.

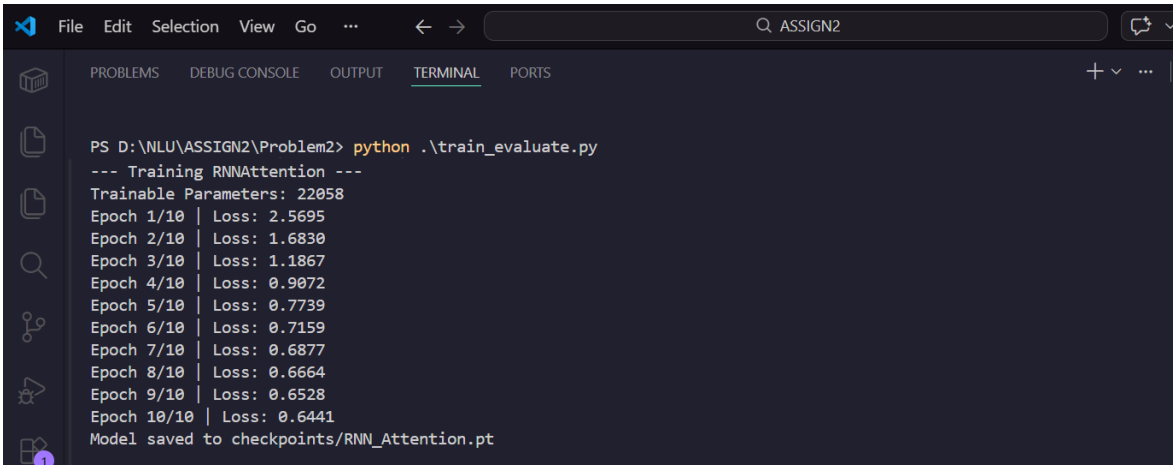


Figure 15: RNN + Attention Loss

2.4 Failure Mode Analysis

Failure Mode	Description	Models
Character Repetition	AAA, PPP patterns	BLSTM
Gibberish Sequences	Unpronounceable combinations	BLSTM
Short Truncation	Incomplete names	Vanilla RNN, Attention
Rare Char Errors	Unusual combinations	All (minor)

Table 8: Common Failure Modes in Name Generation

3 Key Findings and Conclusions

3.1 Problem 1: Word Embeddings

- **Skip-gram superior for semantics:** Tighter clusters, better rare word handling, responsive to hyperparameters. Final loss 3.60 (optimal config).
- **CBOW faster but diffuse:** Stable across parameters, captures co-occurrence but less semantic precision. Final loss 5.67 (optimal config).
- **Best-performing configuration among tested settings:** Dimension=100, Window=6, Negative Samples=10 for Skip-gram. CBOW relatively insensitive.
- **Analogy limitation:** Both models struggle with proportional relationships.
- **Domain learning successful:** Models captured institutional academic structure (degree types, temporal organization, regulations).

3.2 Problem 2: Sequence Generation

- **BLSTM severely overfitted:** Perfect quantitative metrics are artificial and misleading. Output is gibberish.
- **RNN + Attention optimal:** Best balance of quality (realistic names) and metrics (93% diversity, 88% novelty). Attention prevents overfitting.
- **Vanilla RNN reliable baseline:** Good structure and phonetics with genuine generalization (93%/93.55%).
- **Quantitative metrics insufficient:** Diversity/novelty alone miss overfitting. Require qualitative assessment for reliable evaluation.
- **Architecture appropriateness matters:** Simpler models with attention outperformed complex bidirectional models for unidirectional generation.

3.3 Learnings

- Use Skip-gram for semantic tasks requiring fine-grained relationships; CBOW for efficiency-critical applications.
- Prefer RNN + Attention over BLSTM for character-level generation tasks.
- Careful hyperparameter tuning is critical—different architectures respond differently
- Complexity is not always preferable; simpler models sometimes outperform complex ones
- Multiple evaluation metrics (quantitative + qualitative) are essential
- Domain understanding informs preprocessing, architecture choice, and result interpretation

Embeddings capture semantic structure through distributional semantics; RNNs learn sequential patterns. Both are fundamental to modern NLP systems.

3.4 GitHub Repository:

https://github.com/Jyoti-Dwivedi-010/M25CSA010_NLU_A2